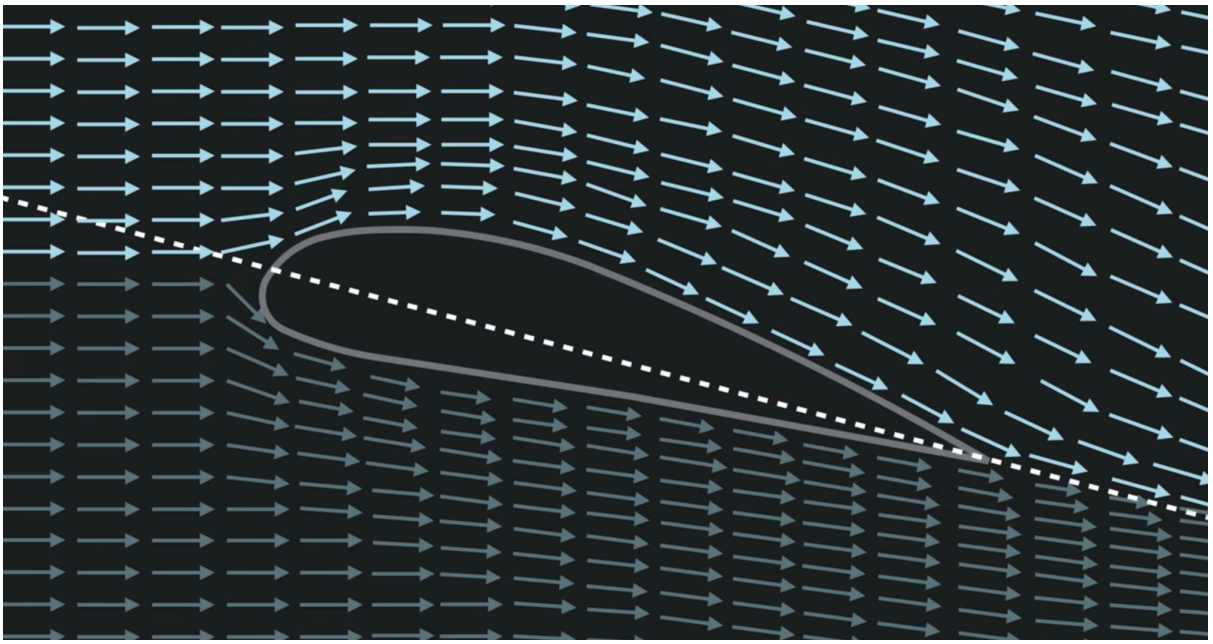
1. Active Hydrofoil Mechanism

a. Find out and explain how a hydrofoil works so it can lift a vessel out of the water, and what's the purpose?

Hydrofoil works by redirecting fluid (water) in a specific way.
https://www.youtube.com/watch?v=fzig8JR2iT0

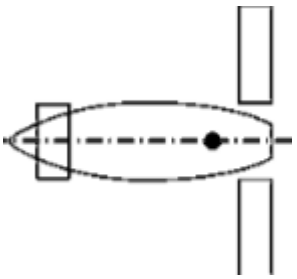


The shape of the hydrofoil redirects the water into two areas, the upper part of the foil and the lower part of the foil. Because of the way gravity works, the water redirected to the lower part increases the pressure (because increasing depth means increasing hydrostatic pressure and water redirected below means the water gets denser) while the water redirected to the upper part decreases the pressure (because the water redirected to the upper part has lower density). This difference in pressure between the lower and upper part generates a lift, hence why a hydrofoil can make a vessel float. At high velocity, this difference in pressure gets higher (Bernoulli principle high speed simultaneously lower pressure) and generates even more lift until the vessel is out of the water.

The purpose of generating this lift until the vessel is out of the water is to reduce drag/resistance. Water is denser than air and therefore generates more drag. By lifting the vessel in the air, the drag gets way lesser.

b. Active hydrofoil allow us to change the foil's angle of attack. What's the benefit? What is the effect of the foil angle on the vessel's height and orientation stability?

By changing the angle of attack, it allows the us to control the roll and pitch orientation of the vessel. This allows the vessel to stay stable even in unfavorable waves. Suppose we have these hydrofoil specification:

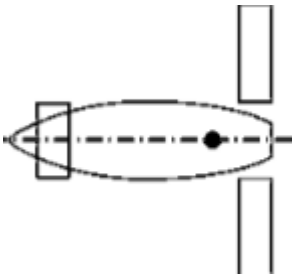


By increasing or decreasing the front and aft foil's angle of attacks, the vessel's pitch can be controlled. The higher the angle of attack, the higher the height and the lower the angle of attack, the lower the height. To control the orientation (roll), the left and right aft foil's angle of attack can be adjusted. if we want to stabilize the vessel to roll to the right, lower the right aft foil's angle of attack and increase the left aft foil's angle of attack and vice versa.

c. The image above shows Jalapatih VI, the latest vessel from ITS MBST, along with its specifications. If you were tasked to designing a hydrofoil for this vessel, which foil profile would you use, and what would the configuration be (regarding quantity,shape, and placement on the vessel)? Please provide your reasoning.



The type of foil used will be a submerged foil.



The foil configuration would be like the above. I chose this configuration (1 foil front and 2 aft foil back at right and left each) because of the relatively small size of the vessel. The small size of the vessel means it prioritizes speed and maneuverability (I assume), which this design very much support. The aft foils makes it easier to stabilize a high moving and volatile steering style while the front provides general pitch support.

2. You are required to control two aspects of your vessel: its height relative to the water surface and its orientation stability. Based on this video and any other sources you have studied, how is the PID controller used to manage these two factors? What effect does adjusting each of the P, I, and D constants have on the vessel? Please provide the complete control block diagram.

Error value: the difference between the target and current position of the vessel.

- Propotional: Produces control output propotional to the error value. The larger the error the larger the control output (higher angle of attack) and vice versa.

- Integration: Produces control output based on the sum of change of the error value. This is particularly useful to avoid steady error where the error stays the same even though the P value already outputs a control value (for example if the target is a moving one, where their speed is the same as the output of our P value).

- Derivative: Produces control output based on the rate of change of the error value. This is useful to stop oversteering or overadjustment. P value outputs are based only on the error value, it does not account for the momentum brought by rapidly adjusting the vessel. To counteract that fast change, D value predicts that rate of change through a derivative and puts more resistance the larger the rate of change, essentially being a stopper/stabilizer of the system.

Because of the need for stability for vessels like jalapatih, the PID also needs to be versatile. The PID controller not only has to stabilize the pitch, but also the roll through the foils. The solution for that problem is through multi input multi output PID system where the input would be from the height of the vessel above the water and the gyroscope sensors to monitor the pitch and roll. From that input, the PID will have multiple outputs, which through the actuators, change the angle of attack of the foils.

